

1. Setting up a bootable device.

For device to boot there is a need for microSD Flash drive.

Another boot option is to use onboard SPI Flash, available in [a variant with 16MB SPI flash memory \(suffix -s16M\)](#) and related u-boot with TFTP kernel load for network. [Howto prepare SPI Flash boot.](#)

Standard microSD Flash has FAT32 file system. It can be accessed from both MS Windows and GNU Linux operating systems. Thus it is not formatted to any linux-specific file system because there are no extra features required to system to work. Files are easily copied to / from the disk.

Getting ARM-based operating system distribution to update binary files of used programs. Or to perform needed setup to created minimal bootable system for CU.

- **Allwinner A20/T2 OLinuXino-LIME2** [download image](#) from [Debian distro](#). There is an Ubuntu option also. [Purchase chip&dip](#), [olimex](#).
- ----

Last Armbian used: Armbian_20.02.1_Lime2_bionic_current_5.4.20

ARM-base minimal OS creation with ~ 8M bytes size and microSD Flash with FAT32 filesystem support.

- 1.1. Need to create **initramfs** inside running ARM GNU/Linux distro **cu-tools/fs-host-armbian/mkinitramfs.sh**. Downloaded image must be installed into Flash, boot, work inside Armbian system. Create file e.g. **cu-tools/fs-boot-disk-SUN7i-olinuxino-lime2/boot/initramfs-***. To run such distro [gemu](#) can be used because all generated files are ARM-device independent (TODO). If init and other scripts will use only few shell tools and dynamic libraries addition or optimization is not needed then filesystem can be also any (TODO).

- 1.2. Then all files can be updated on any usual Linux or other system that supports filesystem with hardlinks/symlinks and `cpio` command. Update examples: binaries like busybox, kernel modules (if used, and newer linux kernel was added). To completely

See:

cu-tools/fs-host-armbian/armbian+usr_share_initramfs-tools/ source and **cu-tools/fs-host-any-gnu-linux/initramfs/** product.

- 1.3. Manual update of existing initramfs image is possible inside Linux-compatible system where file system supports symbolic / hardlinks used.

- 1.3.1 unarchive file from **cu-tools/fs-host-any-gnu-linux/initramfs/initramfs-dir.tar.gz**

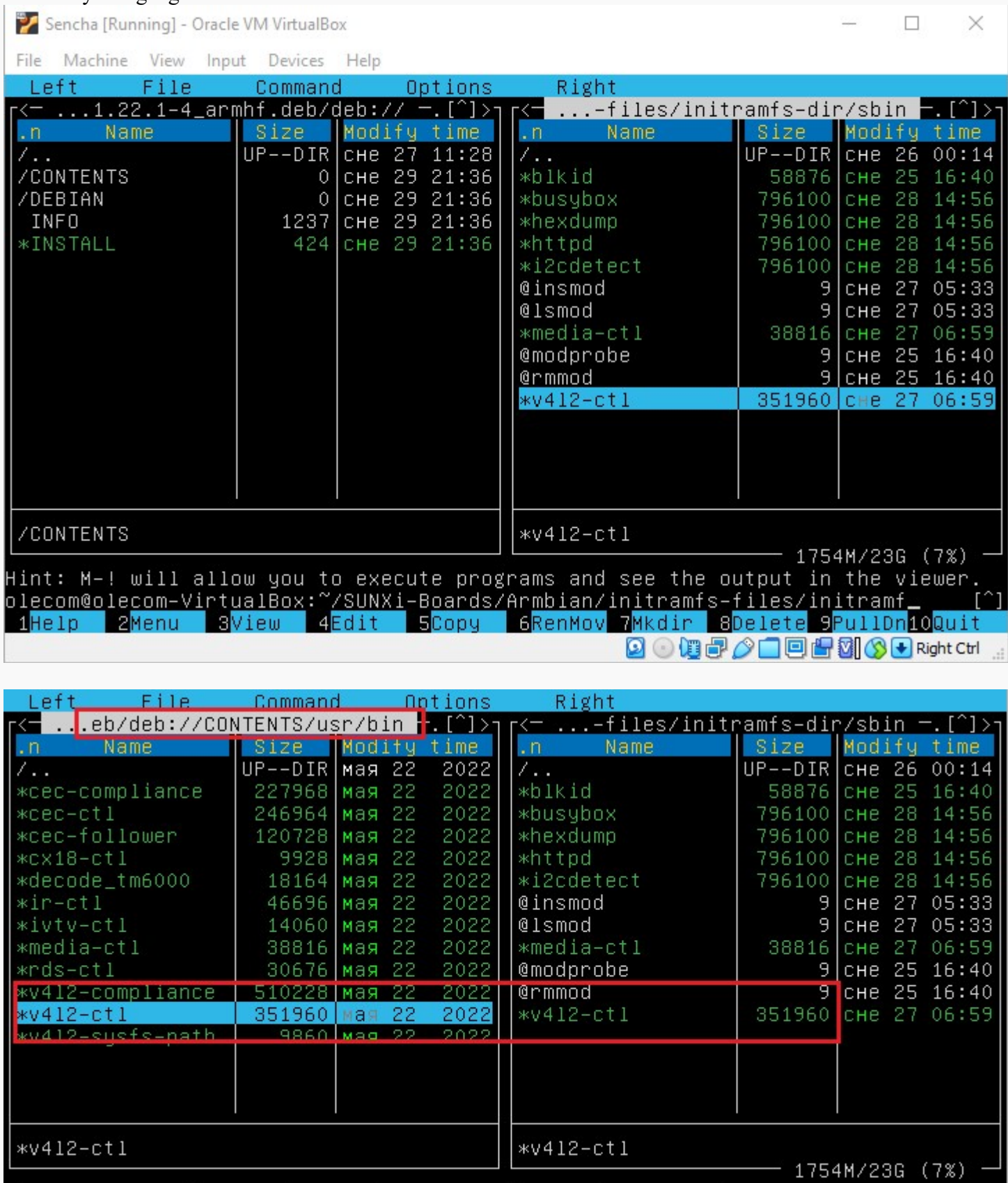
- 1.3.2 download and copy files from distribution files, e.g.

<https://packages.ubuntu.com/jammy/v4l-utils> (with all depended libraries, e.g. <https://packages.ubuntu.com/jammy/libgcc-s1>)

http://ports.ubuntu.com/pool/universe/v/v4l-utils/v4l-utils_1.22.1-4_armhf.deb

http://ports.ubuntu.com/pool/main/v/v4l-utils/libv4l-0_1.22.1-4_armhf.deb

into **initramfs-dir** directory using e.g. **mc** command shell:



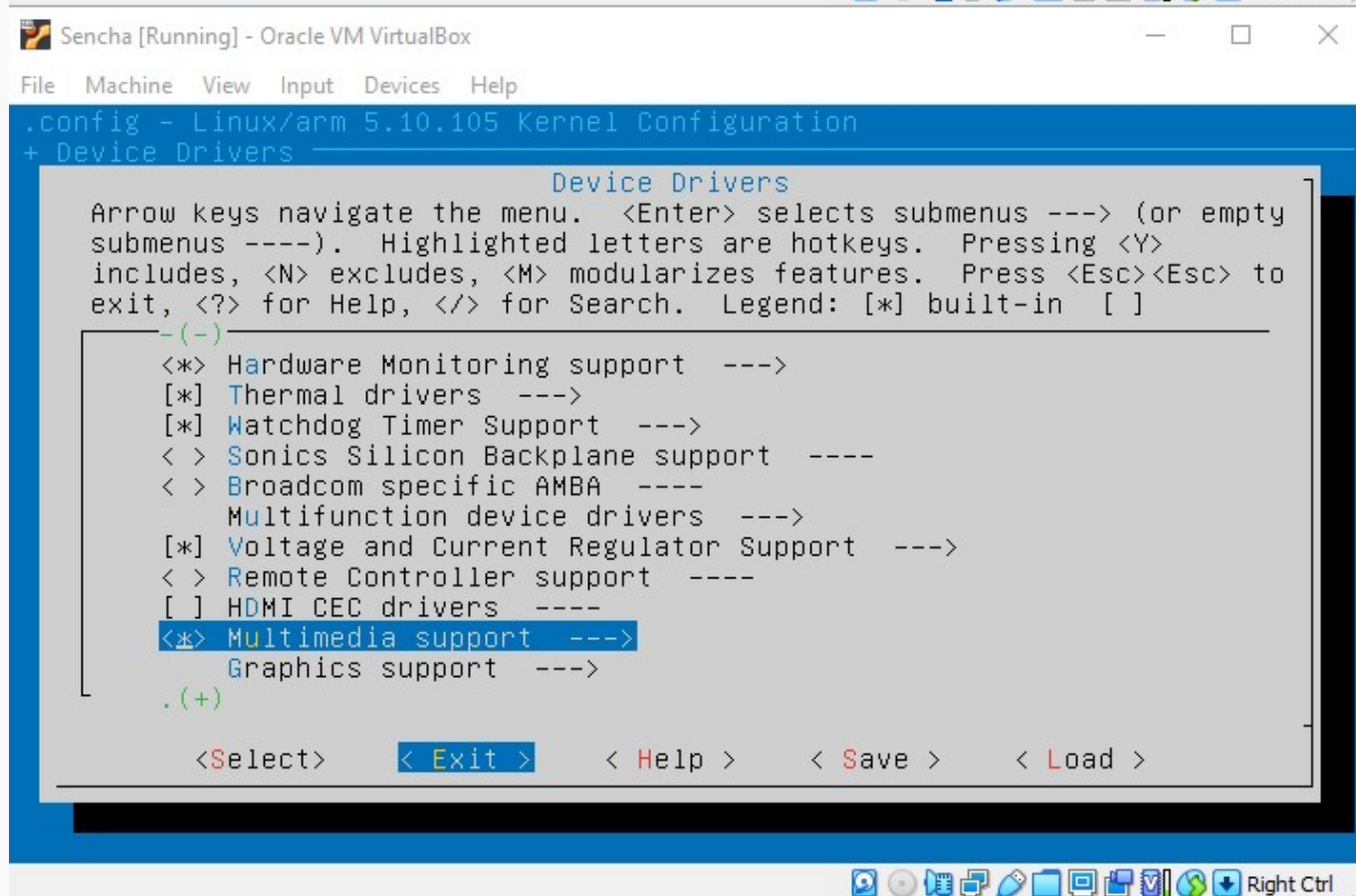
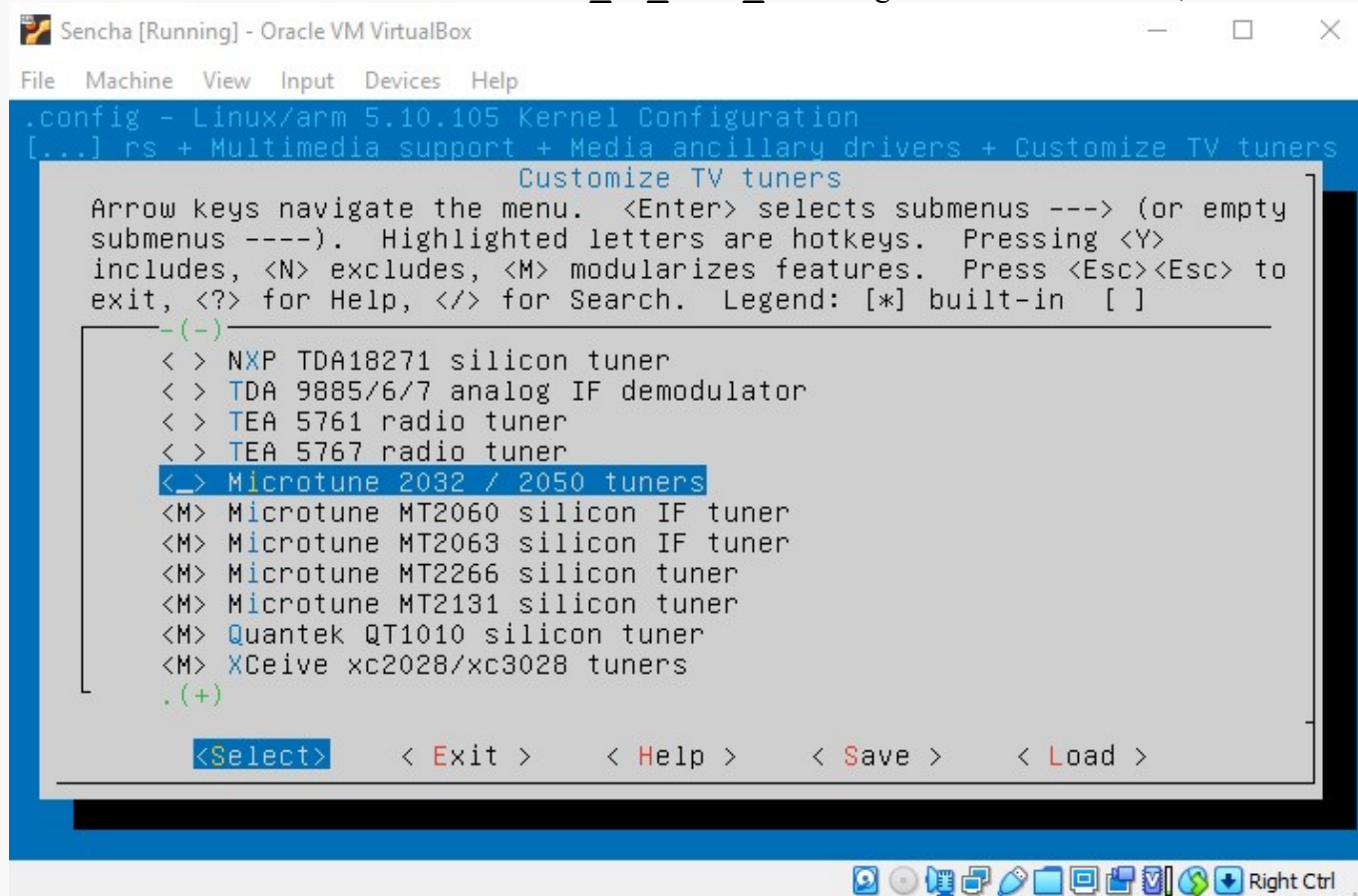
Then generate image using **cu-tools/fs-host-any-gnu-linux/initramfs/mkinitrd.sh** which can be copied into /boot of microFlash card.

2. Compiling Linux kernel with device tree files and Busybox.

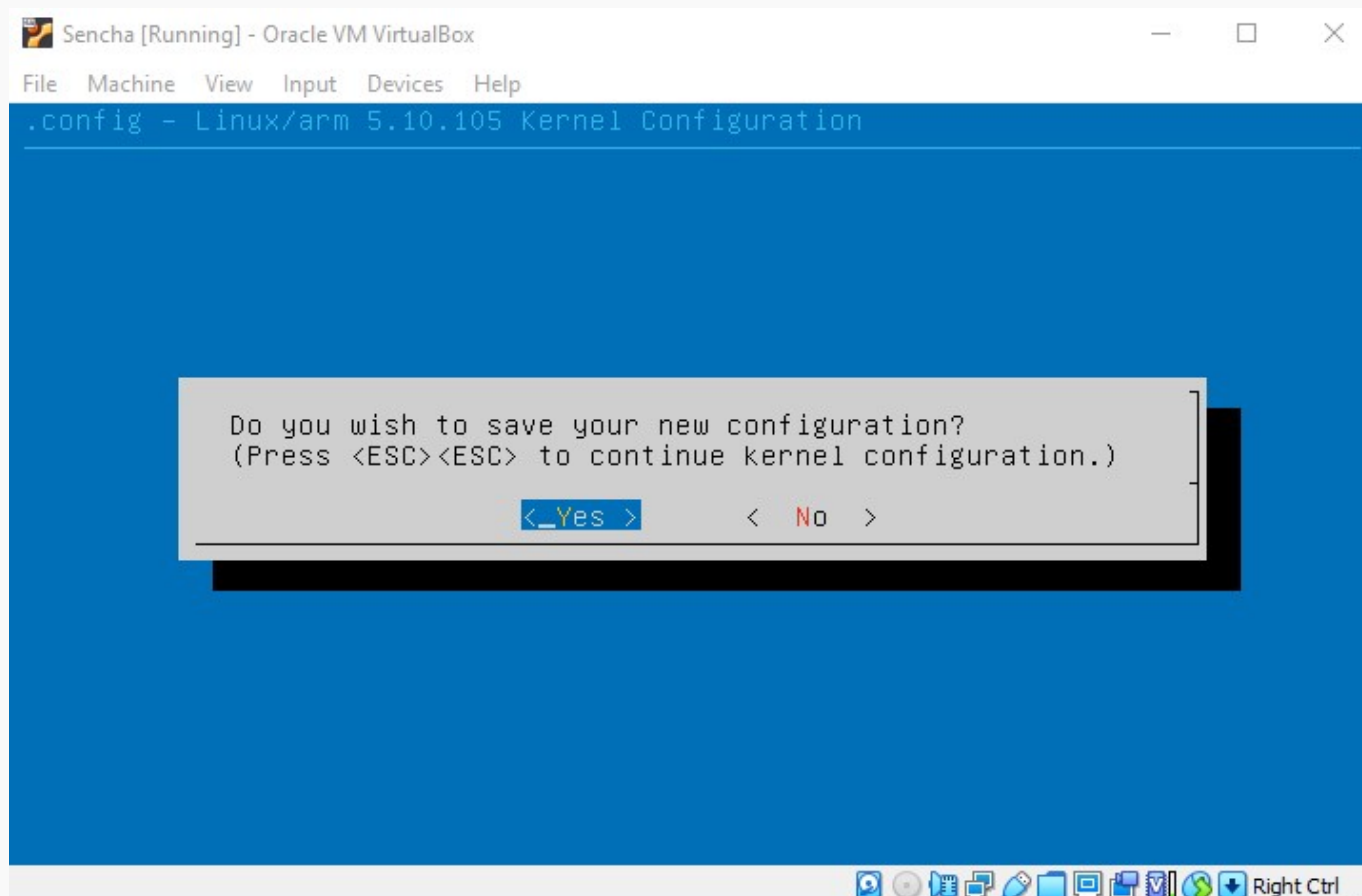
2.1 Get the source: `cu-tools/fs-host-any-gnu-linux/linux-olimax/get-linux-olimax.sh`

2.2 Configure / build script to produce dtb, kernel, modules: `cu-tools/fs-host-any-gnu-linux/linux-olimax/build_linux-olimax.sh`

2.3 Configure. New versions of Linux kernel may have additional features and drivers which are enabled for build. We don't need them. Thus after copying previous `cu-tools/fs-host-any-gnu-linux/linux-olimax/.config_v3_from_v2` config file as new `.config`, all not need item should be removed:

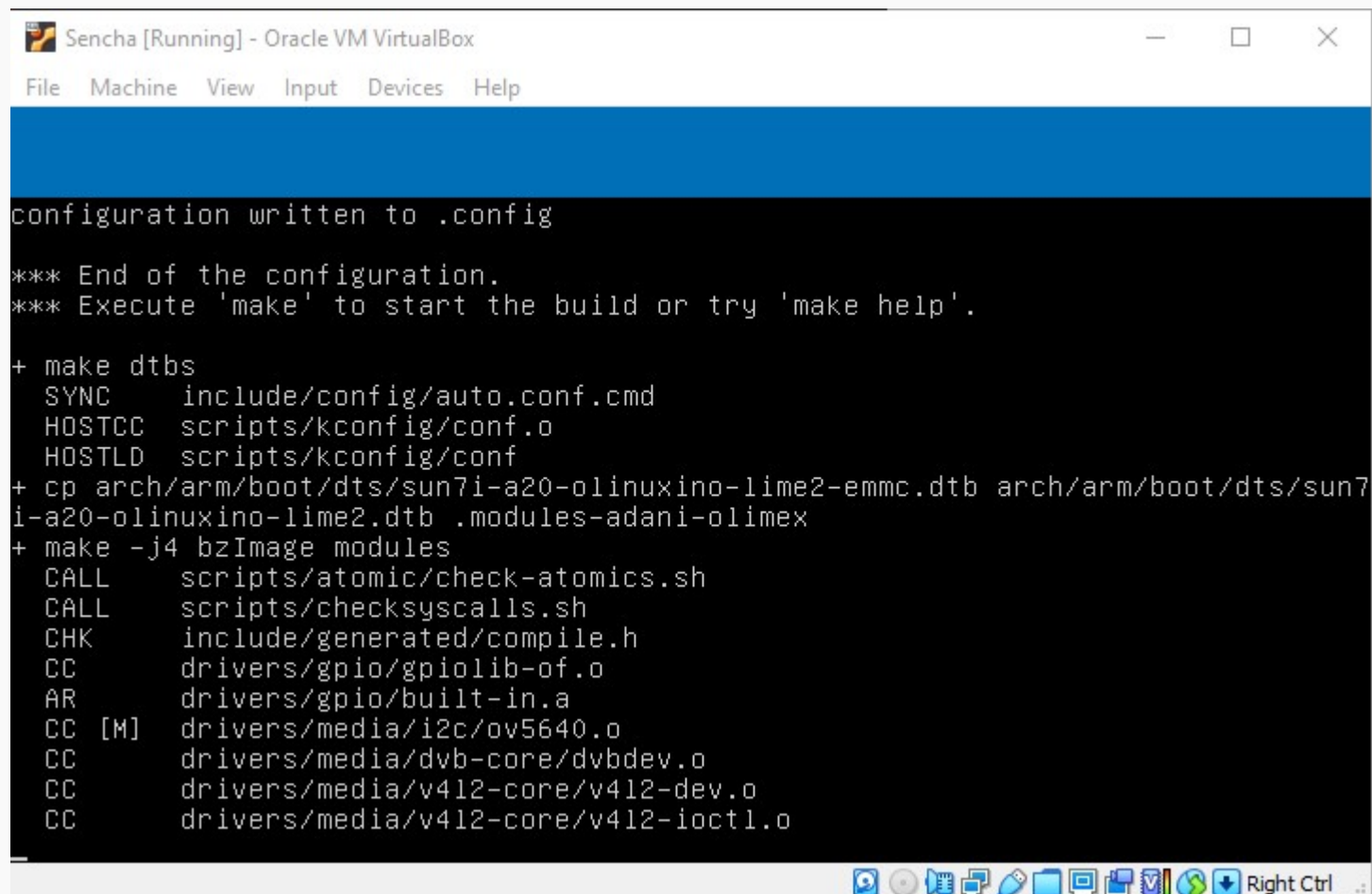


New config file is saved:



2.4 Build happens on usual Linux PC machine (or virtual). Cross compilation into ARM binaries takes place.

Start:



```
configuration written to .config

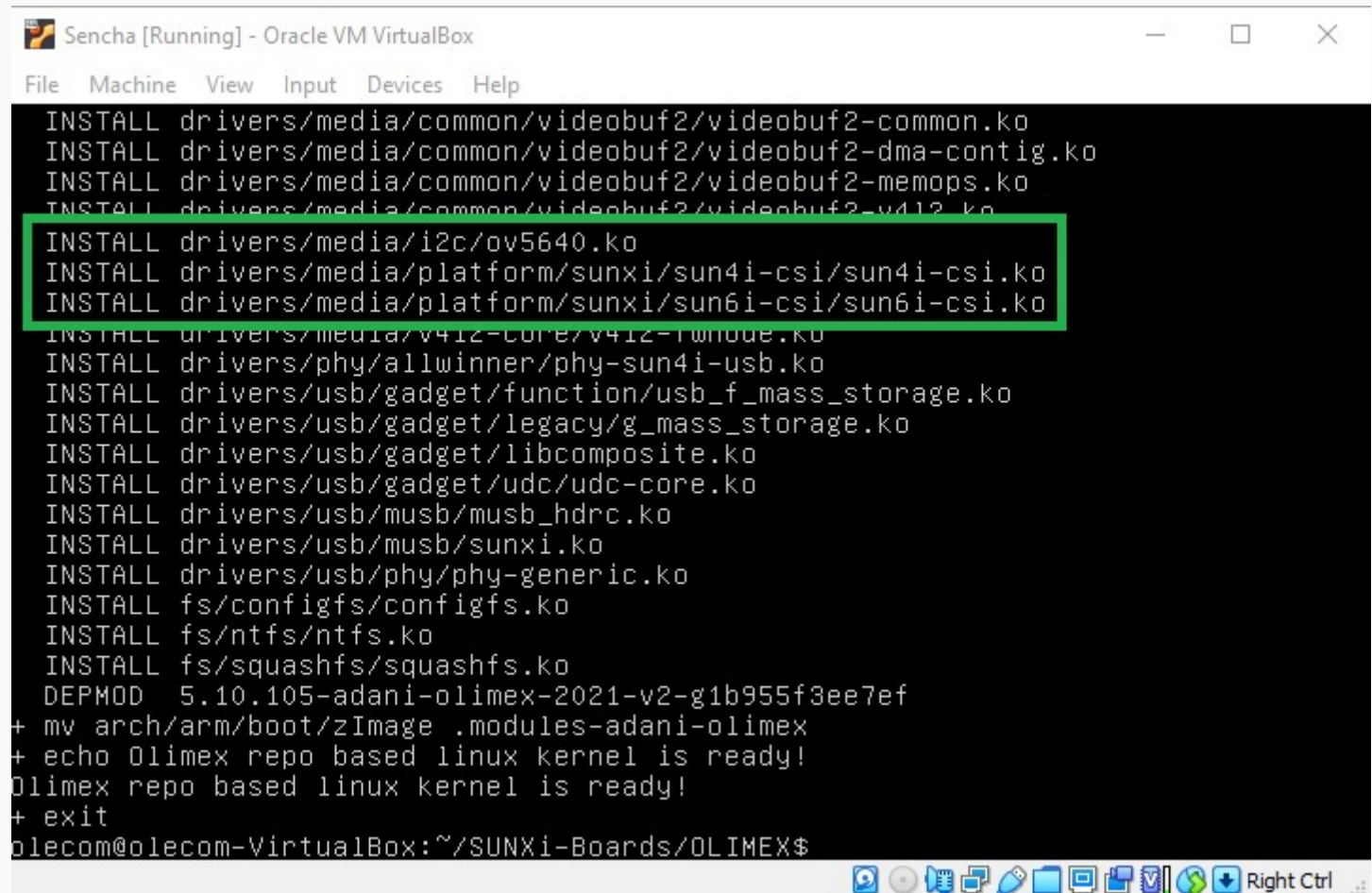
*** End of the configuration.
*** Execute 'make' to start the build or try 'make help'.

+ make dtbs
  SYNC    include/config/auto.conf.cmd
  HOSTCC   scripts/kconfig/conf.o
  HOSTLD   scripts/kconfig/conf
+ cp arch/arm/boot/dts/sun7i-a20-olinuxino-lime2-emmc.dtb arch/arm/boot/dts/sun7i-a20-olinuxino-lime2.dtb .modules-adani-olimax
+ make -j4 bzImage modules
  CALL     scripts/atomic/check-atomics.sh
  CALL     scripts/checksyscalls.sh
  CHK      include/generated/compile.h
  CC       drivers/gpio/gpiolib-of.o
  AR       drivers/gpio/built-in.a
  CC [M]   drivers/media/i2c/ov5640.o
  CC       drivers/media/dvb-core/dvbdev.o
  CC       drivers/media/v4l2-core/v4l2-dev.o
  CC       drivers/media/v4l2-core/v4l2-ioctl.o
```

Finish. Needed driver files are highlighted. Also there are two device tree files built:

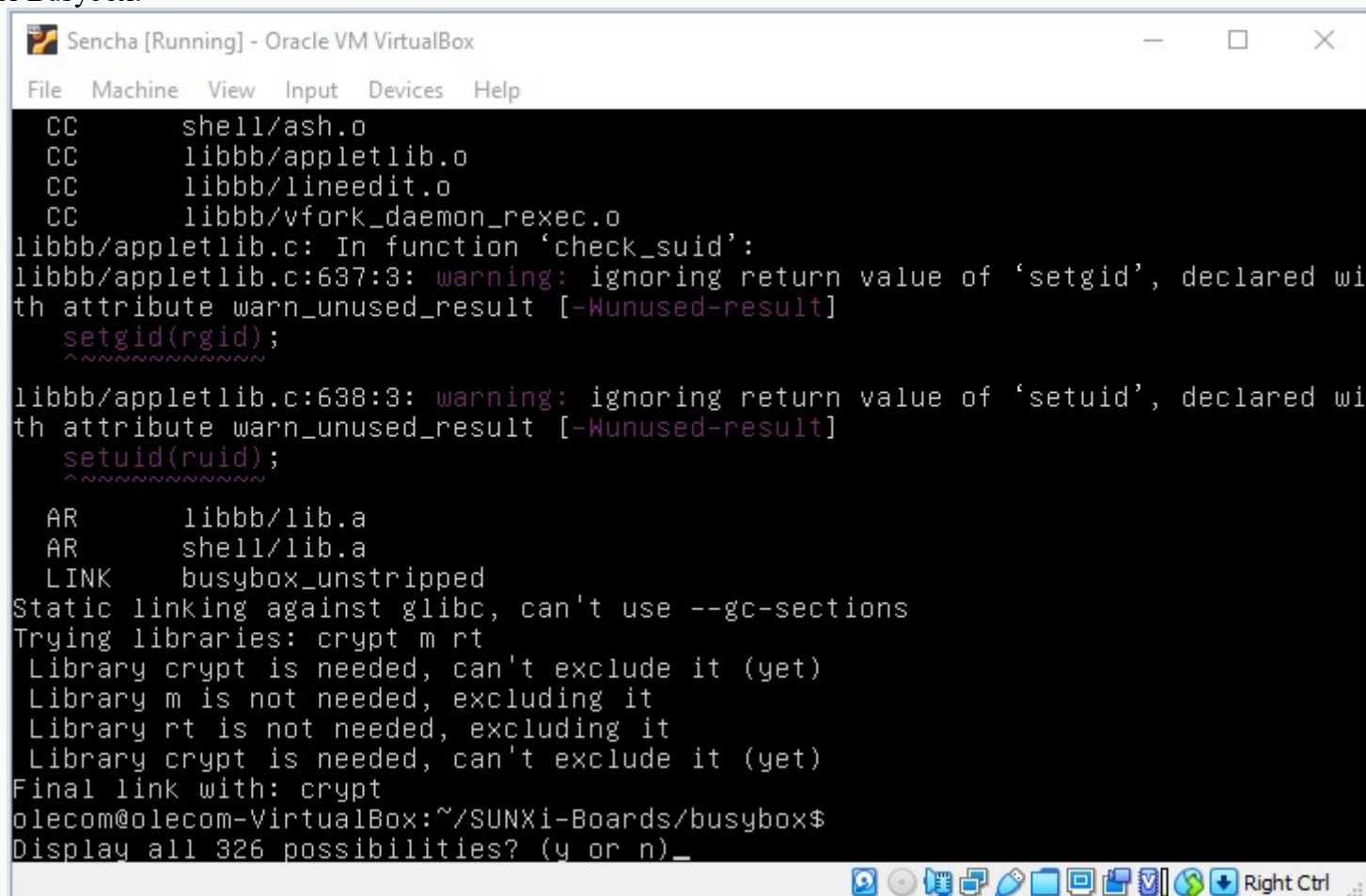
sun7i-a20-olinuxino-lime2.dtb

sun7i-a20-olinuxino-lime2-emmc.dtb



```
INSTALL drivers/media/common/videobuf2/videobuf2-common.ko
INSTALL drivers/media/common/videobuf2/videobuf2-dma-contig.ko
INSTALL drivers/media/common/videobuf2/videobuf2-memops.ko
INSTALL drivers/media/common/videobuf2/videobuf2-v4l2.ko
INSTALL drivers/media/i2c/ov5640.ko
INSTALL drivers/media/platform/sunxi/sun4i-csi/sun4i-csi.ko
INSTALL drivers/media/platform/sunxi/sun6i-csi/sun6i-csi.ko
INSTALL drivers/media/v4l2-core/v4l2-fwnode.ko
INSTALL drivers/phy/allwinner/phy-sun4i-usb.ko
INSTALL drivers/usb/gadget/function/usb_f_mass_storage.ko
INSTALL drivers/usb/gadget/legacy/g_mass_storage.ko
INSTALL drivers/usb/gadget/libcomposite.ko
INSTALL drivers/usb/gadget/udc/udc-core.ko
INSTALL drivers/usb/musb/musb_hdrc.ko
INSTALL drivers/usb/musb/sunxi.ko
INSTALL drivers/usb/phy/phy-generic.ko
INSTALL fs/configfs/configfs.ko
INSTALL fs/ntfs/ntfs.ko
INSTALL fs/squashfs/squashfs.ko
DEPMOD 5.10.105-adani-olimax-2021-v2-g1b955f3ee7ef
+ mv arch/arm/boot/zImage .modules-adani-olimax
+ echo Olimex repo based linux kernel is ready!
Olimex repo based linux kernel is ready!
+ exit
olecom@olecom-VirtualBox:~/SUNXi-Boards/OLIMEX$
```

2.5. Similar build is done for Busybox: <https://github.com/olecom/cu-tools/tree/master/fs-host-any-gnu-linux/busybox>

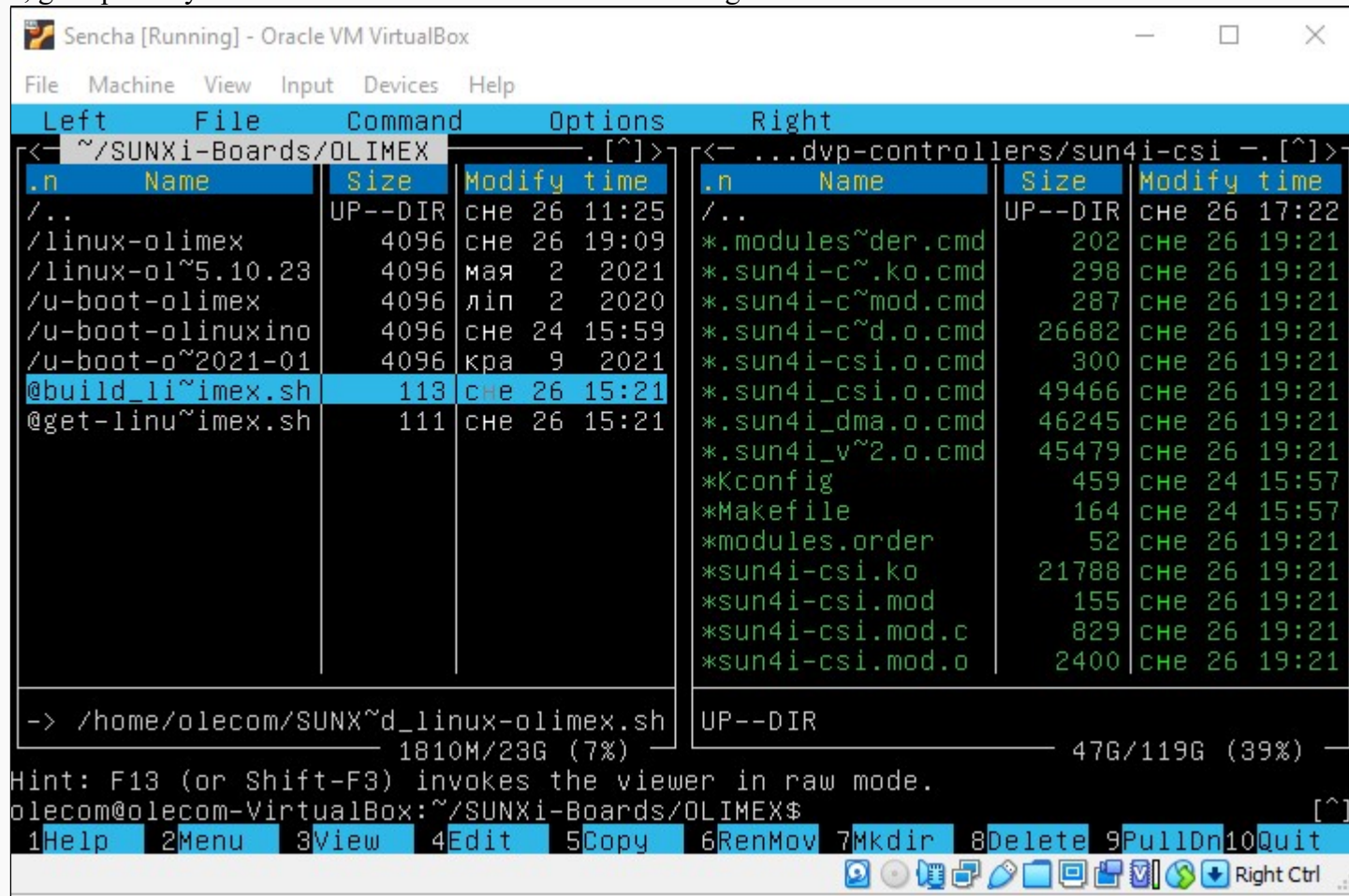


```
CC      shell/ash.o
CC      libbb/appletlib.o
CC      libbb/lineedit.o
CC      libbb/vfork_daemon_rexec.o
libbb/appletlib.c: In function 'check_suid':
libbb/appletlib.c:637:3: warning: ignoring return value of 'setgid', declared with attribute warn_unused_result [-Wunused-result]
  setgid(rgid);
  ~~~~~
libbb/appletlib.c:638:3: warning: ignoring return value of 'setuid', declared with attribute warn_unused_result [-Wunused-result]
  setuid(ruid);
  ~~~~~
AR      libbb/lib.a
AR      shell/lib.a
LINK    busybox_unstripped
Static linking against glibc, can't use --gc-sections
Trying libraries: crypt m rt
Library crypt is needed, can't exclude it (yet)
Library m is not needed, excluding it
Library rt is not needed, excluding it
Library crypt is needed, can't exclude it (yet)
Final link with: crypt
olecom@olecom-VirtualBox:~/SUNXi-Boards/busybox$
Display all 326 possibilities? (y or n)_
```

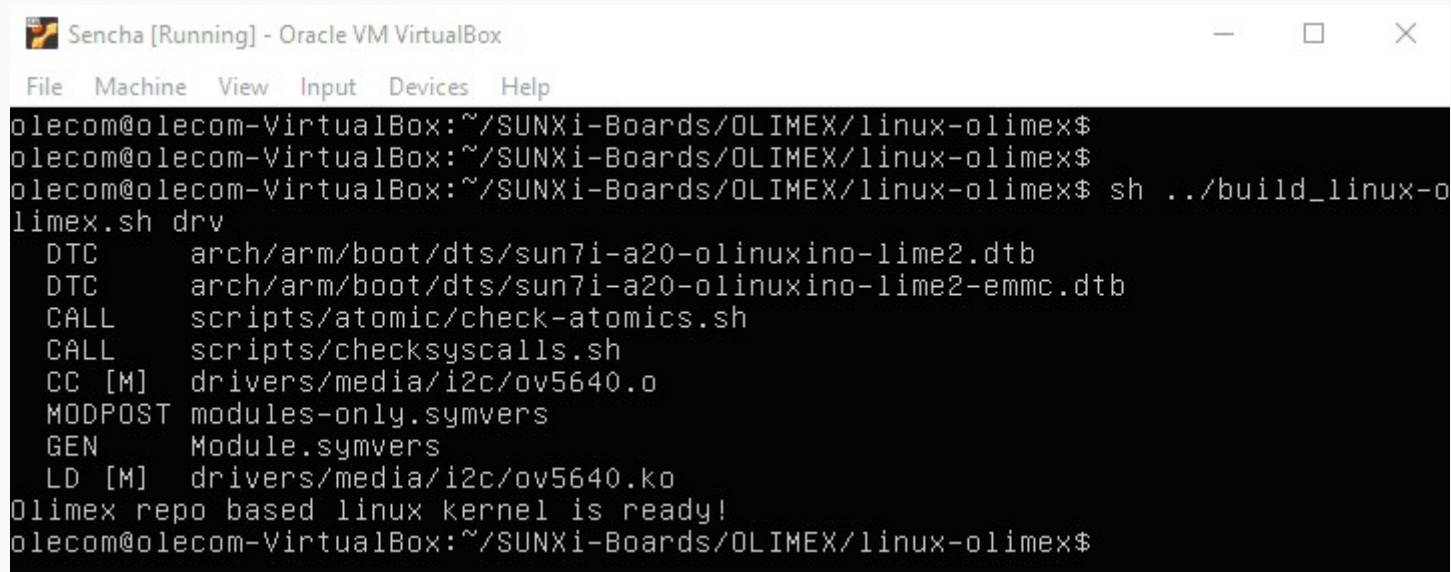
3. Linux kernel drivers development.

Using **cu-tools/fs-host-any-gnu-linux/linux-olimax/build_linux-olimax.sh** source files are connected from **git** on MS Windows (or any other git repository) using symbolic links into linux source tree and then compiled. [Building External Modules](#) is not used, since there is a full linux source available.

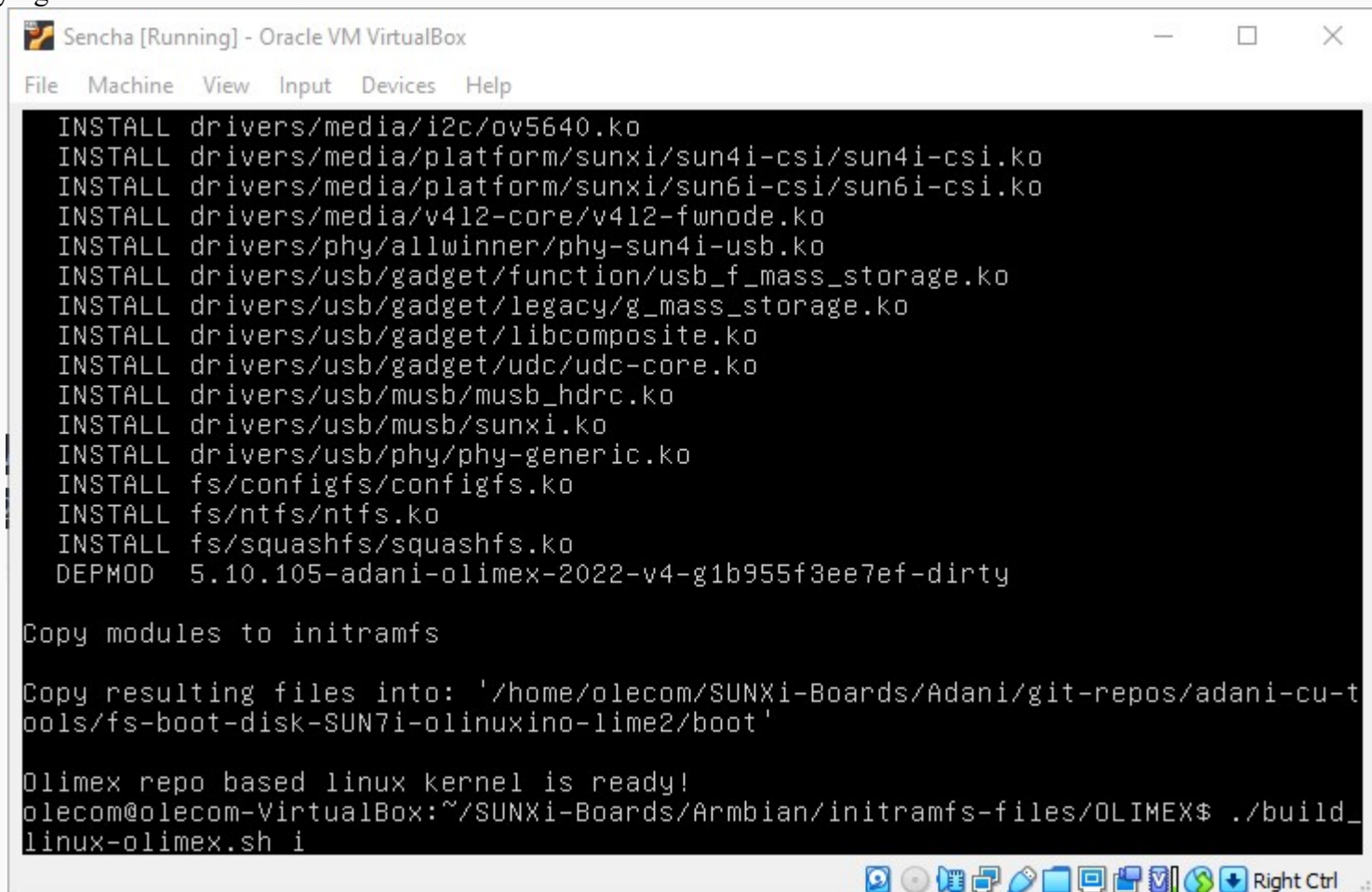
Working directory on the left, git repository **linux-drivers** of CSI drivers on the right:



Compilation process:



Modules installation / copying:



Upload new driver to CU, execute some commands from **cu-arm/exe.sh** with it:

Programming i2c and CSI/DVP drivers using IDE:

linux-drivers - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Team Tools Window Help

Search (Ctrl+F)

ov5640.c sun4i_dma.c sun4i_csi.h sun4i_v4l2.c sun4i_csi.c sun6i_csi.c sun6i_csi_reg.h sun6...

Source History

125 OV5640_FMT_MUX_RGB,
126 OV5640_FMT_MUX_DITHER,
127 OV5640_FMT_MUX_RAW_DPC,
128 OV5640_FMT_MUX_SNR_RAW,
129 OV5640_FMT_MUX_RAW_CIP,
130
131
132 struct ov5640_pixfmt {
133 u32 code;
134 u32 colorspace;
135 };
136
137 static const struct ov5640_pixfmt ov5640_formats[] = {
138 { MEDIA_BUS_FMT_FIXED, V4L2_COLORSPACE_RAW, },
139 { MEDIA_BUS_FMT_JPEG_1X8, V4L2_COLORSPACE_JPEG, },
140 { MEDIA_BUS_FMT_UYVY8_2X8, V4L2_COLORSPACE_SRGB, },
141 { MEDIA_BUS_FMT_YUYV8_2X8, V4L2_COLORSPACE_SRGB, },
142 { MEDIA_BUS_FMT_RGB565_2X8_LE, V4L2_COLORSPACE_SRGB, },
143 { MEDIA_BUS_FMT_RGB565_2X8_BE, V4L2_COLORSPACE_SRGB, },
144 { MEDIA_BUS_FMT_SBGGR8_1X8, V4L2_COLORSPACE_SRGB, },
145 { MEDIA_BUS_FMT_SGBRG8_1X8, V4L2_COLORSPACE_SRGB, },
146 { MEDIA_BUS_FMT_SGRBG8_1X8, V4L2_COLORSPACE_SRGB, },
147 { MEDIA_BUS_FMT_SRGGB8_1X8, V4L2_COLORSPACE_SRGB, },
148 },
149
150 /*
151 * FIXME: remove this when a subdev API becomes available
152 * to set the MIPI CSI-2 virtual channel.
153 */
154 static unsigned int virtual_channel;
155 module_param(virtual_channel, uint, 0444);

SUN7I_LIME_Mammoscan2.c sun4i_dma.c sun4i_v4l2.c

Source History

13 #include <media/v4l2-mc.h>
14 #include <media/videobuf2-v4l2.h>
15
16 #include "sun4i_csi.h"
17
18 #define CSI_DEFAULT_WIDTH 640
19 #define CSI_DEFAULT_HEIGHT 480
20
21 static const struct sun4i_csi_format sun4i_csi_formats[] = {
22 /* X-Ray raw in / out */
23 {
24 .mbus = MEDIA_BUS_FMT_FIXED,
25 .fourcc = V4L2_PIX_FMT_Y16,
26 .input = CSI_INPUT_RAW,
27 .output = CSI_OUTPUT_RAW_PASSTHROUGH,
28 .num_planes = 1, // one FIFO is used for RAW
29 .bpp = { 8, 8, 8 },
30 .hsub = 1, // skip chroma subsampling
31 .vsub = 1, // --
32 },
33 /* YUV422 inputs */
34 {
35 .mbus = MEDIA_BUS_FMT_YUYV8_2X8,
36 .fourcc = V4L2_PIX_FMT_YUV420M,
37 .input = CSI_INPUT_YUV,
38 .output = CSI_OUTPUT_YUV_420_PLANAR,
39 .num_planes = 3,
40 .bpp = { 8, 8, 8 },
41 .hsub = 2,
42 .vsub = 2,
43 },
44 };

Find: CSI_MAX_HEIGHT

Previous Next Select

1 match

Search Results

File

Found 2 matches of CSI_MAX_HEIGHT in 2 files.

sun4i_csi.h

55: #define CSI_MAX_HEIGHT8192U [column 9]

sun4i_v4l2.c

135: pix->height = clamp(height, _fmt->vsub, CSI_MAX_HEIGHT); [column 42]

137:1/2: 102

INS

4. Camera interface info: **sun4i_csi** driver with **sun7i-a20 CSI0** controller, parallel (DVP, DCMI, CSI) interface.

4.1 NOTE: Camera signal name HREF but A20 CSI0 controller has HSYNC name in documentation.

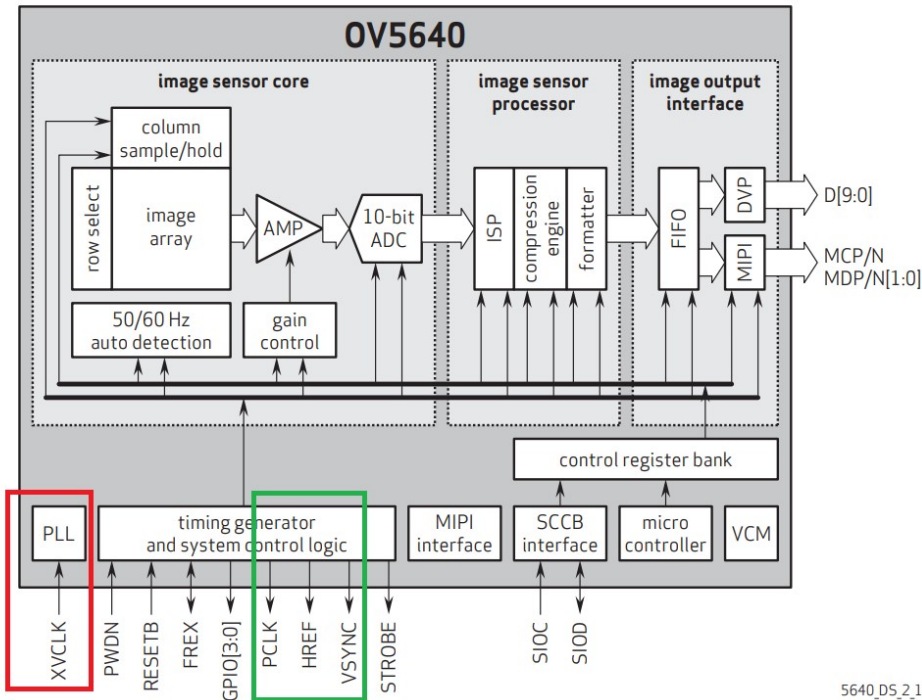
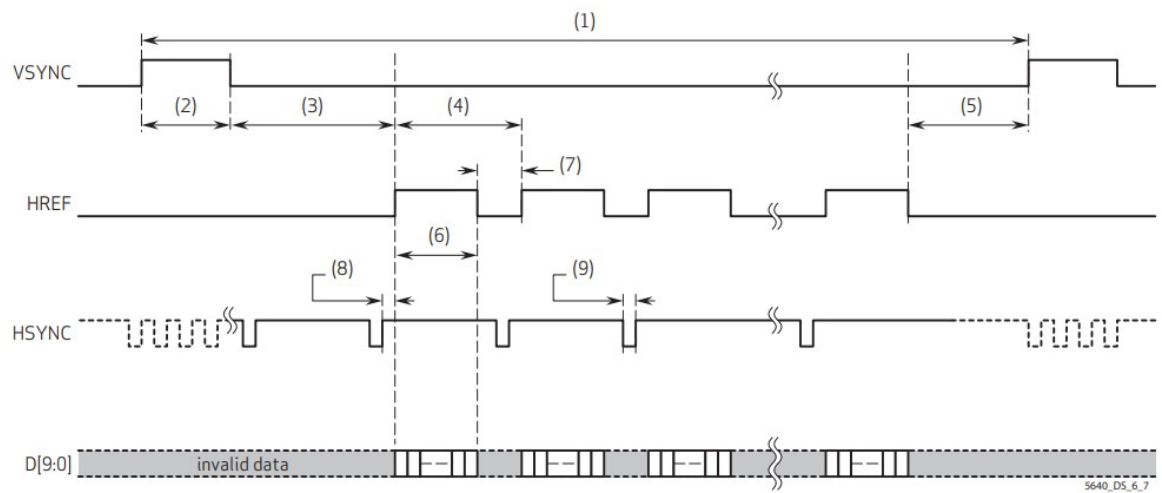


figure 6-7 DVP timing diagram



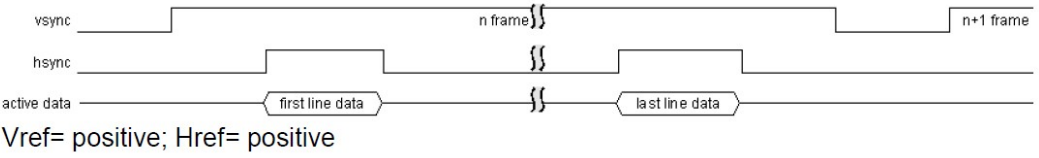
5640_DS_2_1 table 6-7 DVP timing specifications (sheet 1 of 2)

GPIO_2

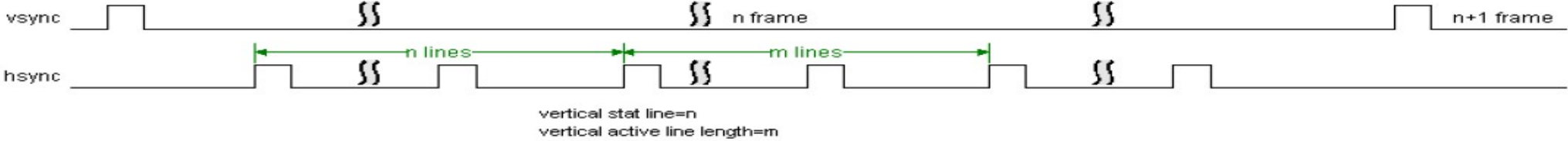
A20 PIN FUNCTIONS	SIGNAL # NAME	#	#	SIGNAL # NAME	A20 PIN FUNCTIONS
-	5V	1	2	GND	-
-	3.3V	3	4	LD03-2.8V	-
TWI0_SCK/PB0	TWI0-SCK	5	6	PE0	TS0_CLK/CSI0_PCLK/PE0 IN
TWI0_SDA/PB1	TWI0-SDA	7	8	PE1	TS0_ERR/CSI0_MCLK/PE1 OUT
GPS_CLK/PI0	PI0	9	10	PE2	TS0_SYNC/CSI0_HSYNC/PE2 IN
GPS_SIGN/PI1	PI1	11	12	PE3	TS0_DVLD/CSI0_VSYNC/PE3 IN
GPS_MAG/PI2	PI2	13	14	PE4	TS0_D0/CSI0_D0/PE4
PWM1/PI3	PI3	15	16	PE5	TS0_D1/CSI0_D1/PE5
SDC3_CMD/PI4	PI4	17	18	PE6	TS0_D2/CSI0_D2/PE6
SDC3_CLK/PI5	PI5	19	20	PE7	TS0_D3/CSI0_D3/PE7
SDC3_D0/PI6	PI6	21	22	PE8	TS0_D4/CSI0_D4/PE8
SDC3_D1/PI7	PI7	23	24	PE9	TS0_D5/CSI0_D5/PE9
SDC3_D2/PI8	PI8	25	26	PE10	TS0_D6/CSI0_D6/PE10
SDC3_D3/PI9	PI9	27	28	PE11	TS0_D7/CSI0_D7/PE11
SPI0_CS0/UART5_TX/EINT22/PI10	PI10	29	30	PI21	HSDA/UART7_RX/PS2_SDA0/PI21
SPI0_CLK/UART5_RX/EINT23/PI11	PI11	31	32	PI20	HSCL/UART7_TX/PS2_SCK0/PI20
SPI0_MOSI/UART6_TX/EINT24/PI12	PI12	33	34	PI19	EINT31/SPI1_MISO/UART2_RX/PI19
SPI0_MISO/UART6_RX/EINT25/PI13	PI13	35	36	PI18	EINT30/SPI1_MOSI/UART2_TX/PI18
PS2_SCK1/TCLKIN0/EINT26/SPI0_CS1/PI14	PI14	37	38	PI17	EINT29/SPI1_CLK/UART2_CTS/PI17
PS2_SDA1/TCLKIN1/EINT27/SPI1_CS1/PI15	PI15	39	40	PI16	EINT28/SPI1_CS0/UART2_RTS/PI16

4.1.3.2. TIMING DIAGRAM

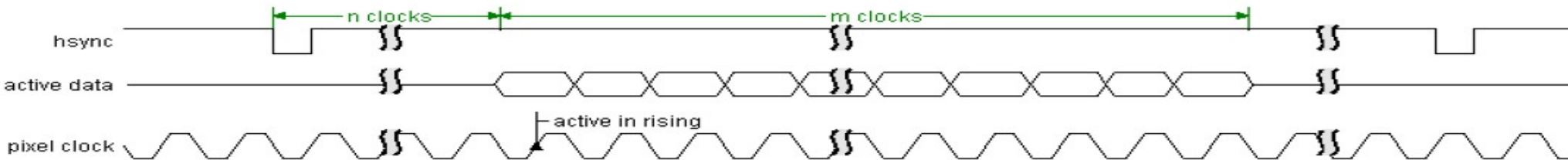
CSI timing



Vref= positive; Href= positive



vertical size setting



OV7670/OV7171 CMOS VGA (OmniPixel®) CAMERACHIP™

Figure 11 RGB 565 Output Timing Diagram

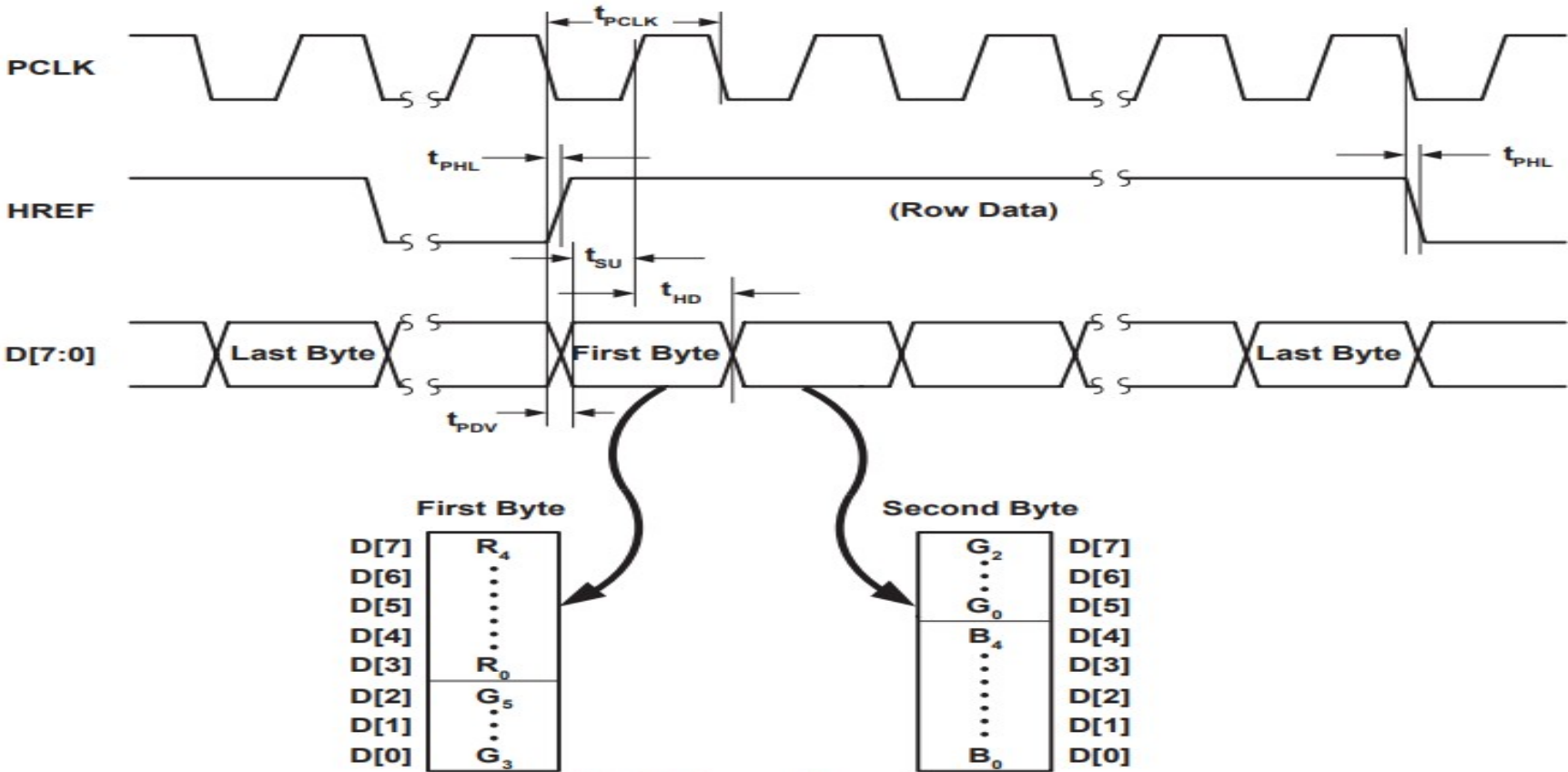


Figure 5 Horizontal Timing

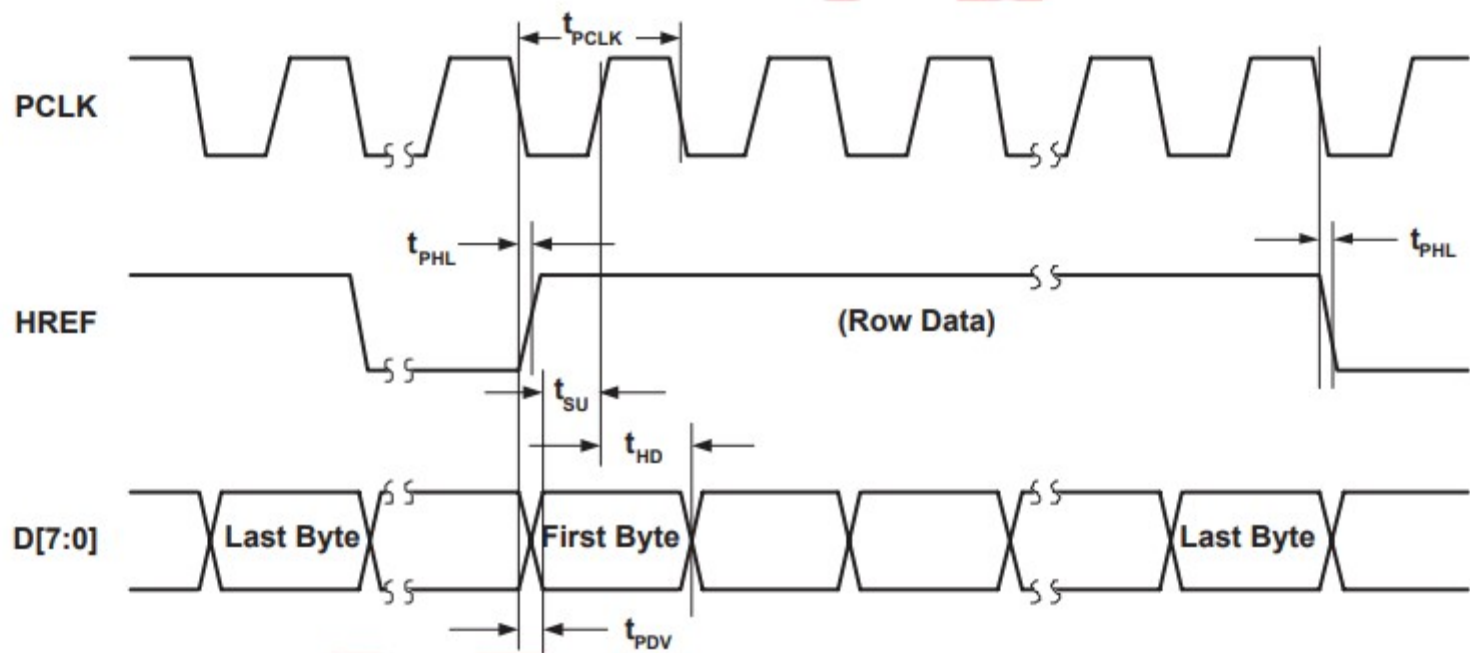
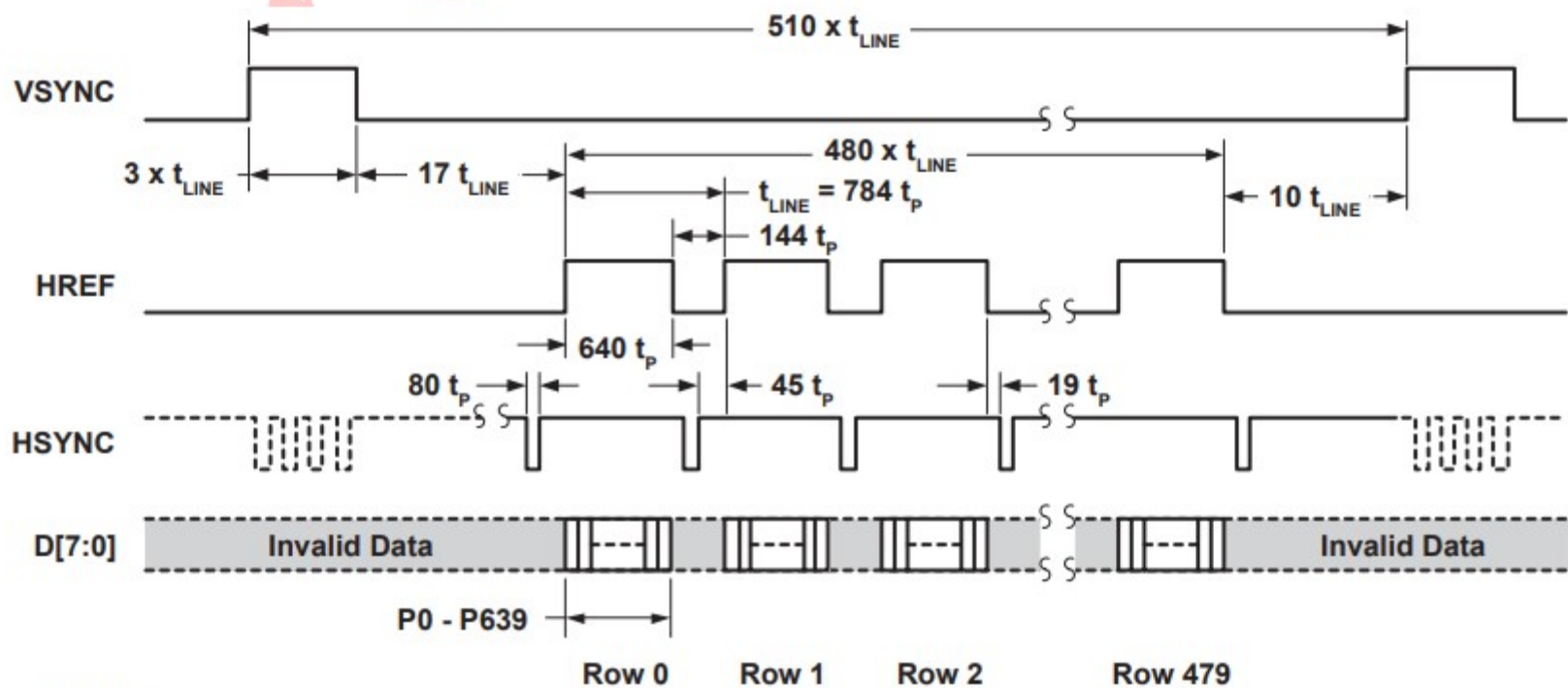


Figure 6 VGA Frame Timing



NOTE:
For Raw data, $t_p = t_{PCLK}$
For YUV/RGB, $t_p = 2 \times t_{PCLK}$

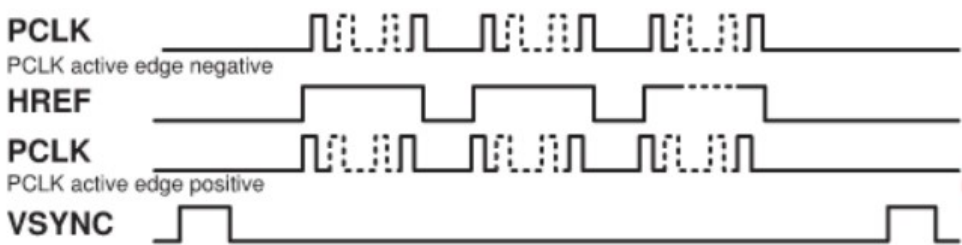
Line/Pixel Timing Ov2640 Camera PDF

The OV2640 digital video port can be programmed to work in either master or slave mode.

In both master and slave modes, pixel data output is synchronous with PCLK (or MCLK if port is a slave), HREF, and VSYNC. The default PCLK edge for valid data is the negative edge but may be programmed using register COM10[4] for the positive edge. Basic line/pixel output timing and pixel timing specifications are shown in Figure 14 and Table 10.

Also, using register COM10[5], PCLK output can be gated by the active video period defined by the HREF signal. See Figure 11 for details.

Figure 11 PCLK Output Only at Valid Pixels



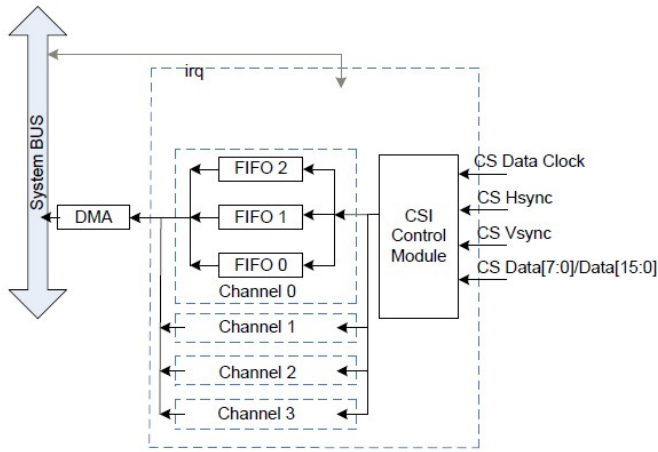
The specifications shown in Table 10 apply for DVDD = +1.2 V, DOVDD = +2.8 V, $T_A = 25^\circ\text{C}$, sensor working at 15 fps, external loading = 20 pF.

4.1. CSI0

4.1.1. Overview

- CSI0 features:
- 8 bits input data
 - Support CCIR656 protocol for NTSC and PAL
 - 3 parallel data paths for image stream parsing
 - Received data double buffer support
 - Parsing bayer data into planar R, G, B output to memory
 - Parsing interlaced data into planar or tiled Y, Cb, Cr output to memory
 - Pass raw data direct to memory
 - All data transmit timing can be adjusted by software
 - Support multi-channel ITU-R BT656 time-multiplexed format
 - Luminance statistical value
 - Support 8-bit raw data input
 - Support 16-bit YUV422 data input

4.1.2. CSI0 Block Diagram



4.1.3. CSI0 Description

4.1.3.1. CSI DATA PORTS

	Bayer	YCbCr (YUV)	Interlaced	Pass-through
FIFO0	Red pixel data	Y pixel data	All field 1 pixel data	All pixel data
FIFO1	Green pixel data	Cb (U) pixel data	All field 2 pixel data	-
FIFO2	Blue pixel data	Cr (V) pixel data	-	-

Question about HREF vs HSYNC from linux driver:

```
/*
 * This hardware uses [HV]REF instead of [HV]SYNC. Based on the
 * provided timing diagrams in the manual, positive polarity
 * equals active high [HV]REF.
 *
 * When the back porch is 0, [HV]REF is more or less equivalent
 * to [HV]SYNC inverted.
 */
href_pol = !(bus->flags & V4L2_MBUS_HSYNC_ACTIVE_LOW);
vref_pol = !(bus->flags & V4L2_MBUS_VSYNC_ACTIVE_LOW);
pclk_pol = !(bus->flags & V4L2_MBUS_PCLK_SAMPLE_RISING);
writel(CSI_CFG_INPUT_FMT(csi_fmt->input) |
       CSI_CFG_OUTPUT_FMT(csi_fmt->output) |
       CSI_CFG_VREF_POL(vref_pol) |
       CSI_CFG_HREF_POL(href_pol) |
       CSI_CFG_PCLK_POL(pclk_pol),
       csi->regs + CSI_CFG_REG);
```

4.3 CSI Clock information: default is 24MHz = OSC24M = 0b000

4.3.1 A20 datasheet information.

1.5.3. CCU Register List

Module Name	Base Address
CCU	0x01C20000

Register Name	Offset	Description
PLL3_CFG_REG	0x0010	PLL3 CONTROL
PLL4_CFG_REG	0x0018	PLL4 CONTROL
PLL5_CFG_REG	0x0020	PLL5 CONTROL
PLL5_TUN_REG	0x0024	PLL5 TUNING
PLL6_CFG_REG	0x0028	PLL6 CONTROL
PLL6_TUN_REG	0x002C	PLL6 TUNING
PLL7_CFG_REG	0x0030	PLL7 CONTROL
/	0x0034	/
CSI_SCLK_REG	0x0120	CSI SPECIAL CLOCK REGISTER
CSI0_CLK_REG	0x0134	CSI0 CLOCK REGISTER
CSI1_CLK_REG	0x0138	CSI1 CLOCK REGISTER

1.5.4.55. CSI 0 CLOCK(DEFAULT: 0X00000000)

Offset: 0x134			Register Name: CSI0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SCLK_GATING. Gating Special Clock 0: Clock is OFF 1: Clock is ON This special clock = Clock Source/Divider M.
30	R/W	0x0	CSI0_RST. CSI0 Reset. 0: reset valid, 1: reset invalid.
29:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL. Clock Source Select
			000: OSC24M 001: PLL3(1X) 010: PLL7(1X) 011: / 100: / 101: PLL3(2X) 110: PLL7(2X) 111: /
23:5	/	/	/
4:0	/	/	CLK_DIV_RATIO_M. Clock divide ratio (m) The pre-divided clock is divided by (m+1). The divider is from 1 to 32.

1.5.4.11. PLL7-VIDEO 1(DEFAULT: 0X0010D063)

Offset: 0x30		Register Name: PLL7_CFG_REG	
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL7_Enable. 0: Disable, 1: Enable. In the integer mode, The PLL7 output=3MHz*M. In the fractional mode, the PLL7 output is select by bit 14. The PLL7 output range is 27MHz~381MHz.
30:16	/	/	/.
15	R/W	0x1	PLL7_MODE_SEL. PLL7 mode select. 0: fractional mode, 1: integer mode.
14	R/W	0x1	PLL7_FRAC_SET. PLL7 fractional setting. 0: 270MHz, 1: 297MHz.
13:7	/	/	/.
6:0	R/W	0x63	PLL7_FACTOR_M. PLL7 Factor M. The range is from 9 to 127.

4.3.2 Linux kernel / driver programming / setup of PCLK frequency:

```
csi0: csi@1c09000 {
    compatible = "allwinner,sun7i-a20-csi0";
    reg = <0x01c09000 0x1000>;
    interrupts = <GIC_SPI 42 IRQ_TYPE_LEVEL_HIGH>;
    clocks = <&ccu CLK_AHB_CSI0>, &ccu CLK_CSI_SCLK, <&ccu CLK_DRAM_CSI0>, <&ccu CLK_CSI0>;
    clock-names = "bus", "isp", "ram", "mclk";
    resets = <&ccu RST_CSI0>;
    status = "disabled";
};
```

```
#define CLK_CSI_SCLK      151 [CLK_CSI_SCLK]      = &csi_sclk_clk.common.hw,
#define CLK_CSI0         158 [CLK_CSI0]          = &csi0_clk.common.hw,
#define CLK_CSI1         159 [CLK_CSI1]          = &csi1_clk.common.hw,
```

Added name for PLL7:

```
#define CLK_PLL_VIDEO1    17
```

```
diff --git a/include/dt-bindings/clock/sun4i-a10-ccu.h
ck/sun4i-a10-ccu.h
index e4fa61be5..1b149973e 100644
--- a/include/dt-bindings/clock/sun4i-a10-ccu.h
+++ b/include/dt-bindings/clock/sun4i-a10-ccu.h
@@ -44,6 +44,7 @@
 
 #define CLK_HOSC          1
 #define CLK_PLL_VIDEO0_2X 9
+#define CLK_PLL7_VIDEO1   17
 #define CLK_PLL_VIDEO1_2X 18
 #define CLK_CPU           20
```

Not MCLK relevant “isp” clock setup:

```
static const char *const csi_sclk_parents[] = { "pll-video0", "pll-ve",
                                                "pll-ddr-other", "pll-periph" };

static SUNXI_CCU_M_WITH_MUX_GATE(csi_sclk_clk, "csi-sclk",
                                csi_sclk_parents,
                                0x120, 0, 4, 24, 2, BIT(31), 0);
```

MCLK relevant:

```
static const char *const csi_parents[] = { "hosc", "pll-video0", "pll-video1",
                                            "pll-video0-2x", "pll-video1-2x" };
static const u8 csi_table[] = { 0, 1, 2, 5, 6 };
static SUNXI_CCU_M_WITH_MUX_GATE(csi0_clk, "csi0",
                                csi_parents, csi_table,
                                0x134, 0, 5, 24, 3, BIT(31), 0);
```

This is PLL7 setup:

```
static struct ccu_mult pll_video1_clk = {
    .enable = BIT(31),
    .mult = _SUNXI_CCU_MULT_OFFSET_MIN_MAX(0, 7, 0, 9, 127),
    .frac = _SUNXI_CCU_FRAC(BIT(15), BIT(14),
                            270000000, 297000000),
    .common = {
        .reg = 0x030,
        .features = (CCU_FEATURE_FRACTIONAL |
                     CCU_FEATURE_ALL_PREDIV),
        .prediv = 8,
        .hw.init = CLK_HW_INIT("pll-video1",
                                "hosc",
                                &ccu_mult_ops,
                                0),
    },
};
```

To change PCLK frequency this way can be used to change file

```
sun7i-a20-olinuxino-lime2.dts file: ov5640: camera@3c {
    compatible = "ovti,ov5640";
    reg = <0x3c>;
    pinctrl-names = "default";
    pinctrl-0 = <&csi0_clk_pin>;
    clocks = <&ccu CLK_CSI0>;
    clock-names = "xclk";
```

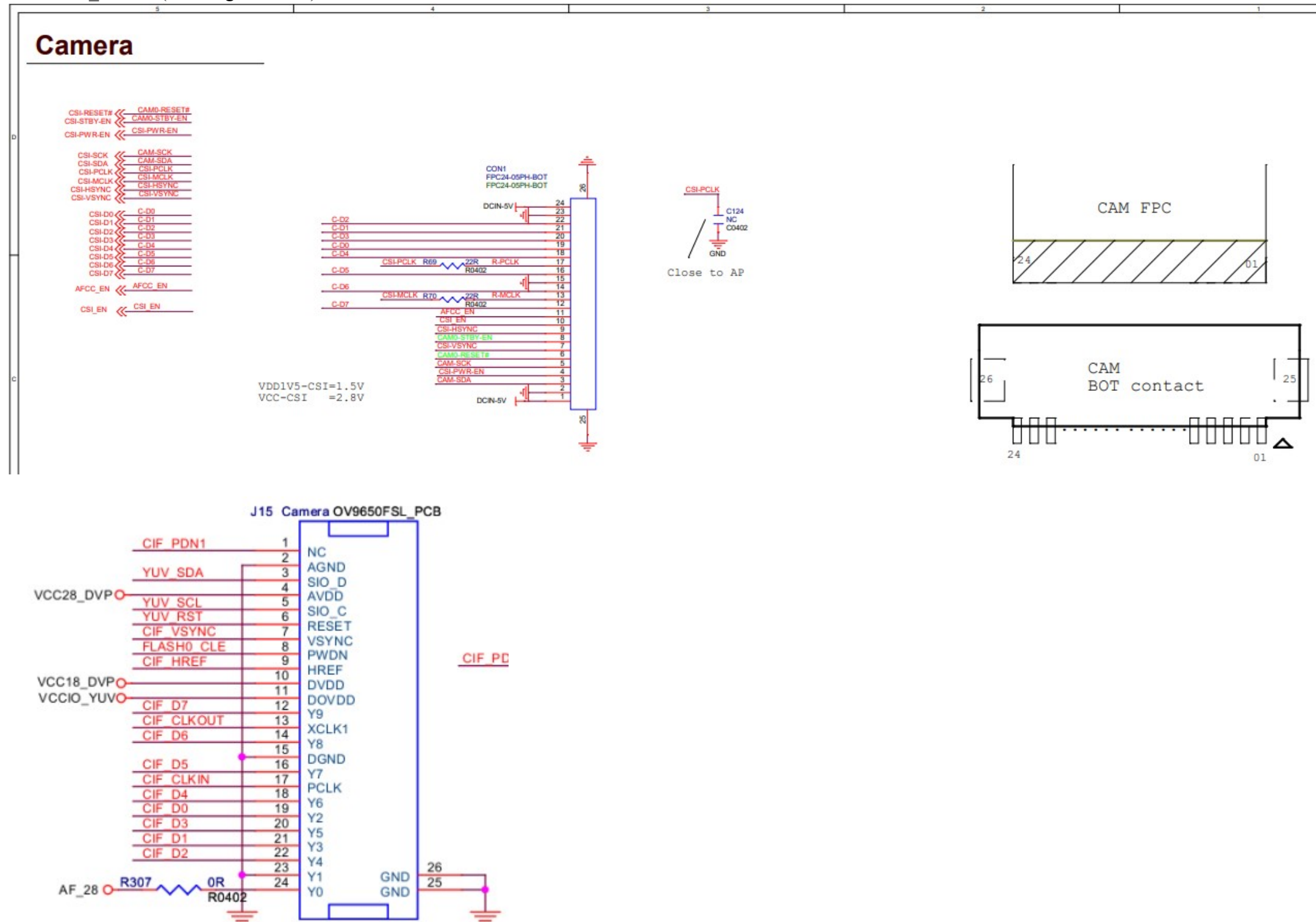
```
/* modify default clock sources / frequencies */
assigned-clocks = <&ccu CLK_CSI0>, <&ccu CLK_PLL7_VIDEO1>;
/* select CSI0 source clock PLL7 (=17) in CSI0_CLK_REG */
assigned-clock-parents = <&ccu CLK_PLL7_VIDEO1>;
/*
 * the integer mode of PLL7: (3 MHz * (9 + 7)) = 48 MHz
 * set CSI0 clock in CSI0_CLK_REG as division by (2 + 1) of the
 * source: 48 MHz / 3 = 16 MHz
 */
assigned-clock-rates = <16000000>, <48000000>;

/* default main oscillator setup: 24000000 Hz
assigned-clocks = <&ccu CLK_CSI0>, <&ccu CLK_HOSC>;
assigned-clock-parents = <&ccu CLK_HOSC>;
*/

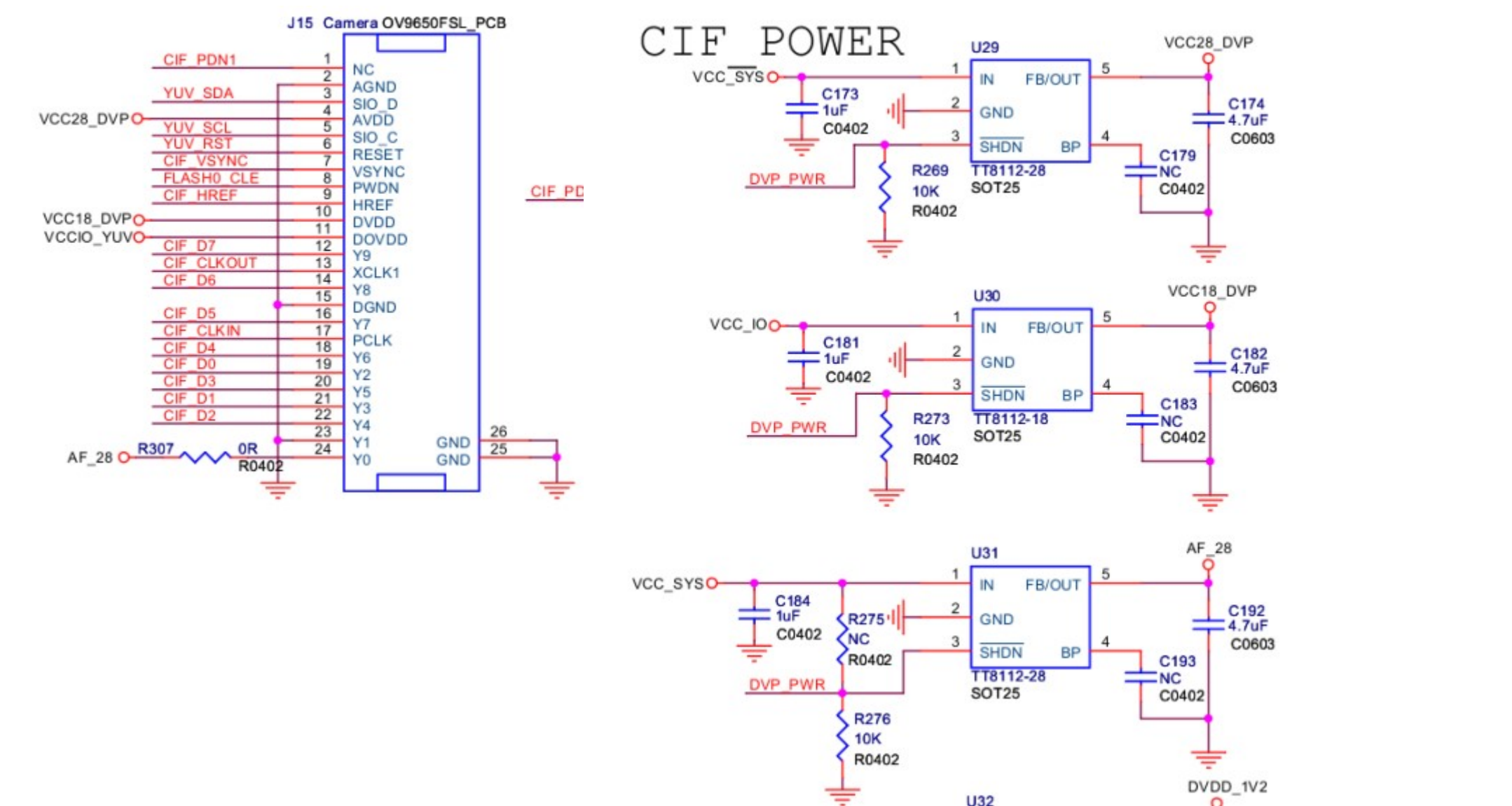
assigned-clock-rates = <16500000>;
```

5. Schematics of DVP parallel camera connector

ORANGE_PI-One (Xunlong Software):



DVP camera interface on the **Firefly-RK3288-Reload** development board:

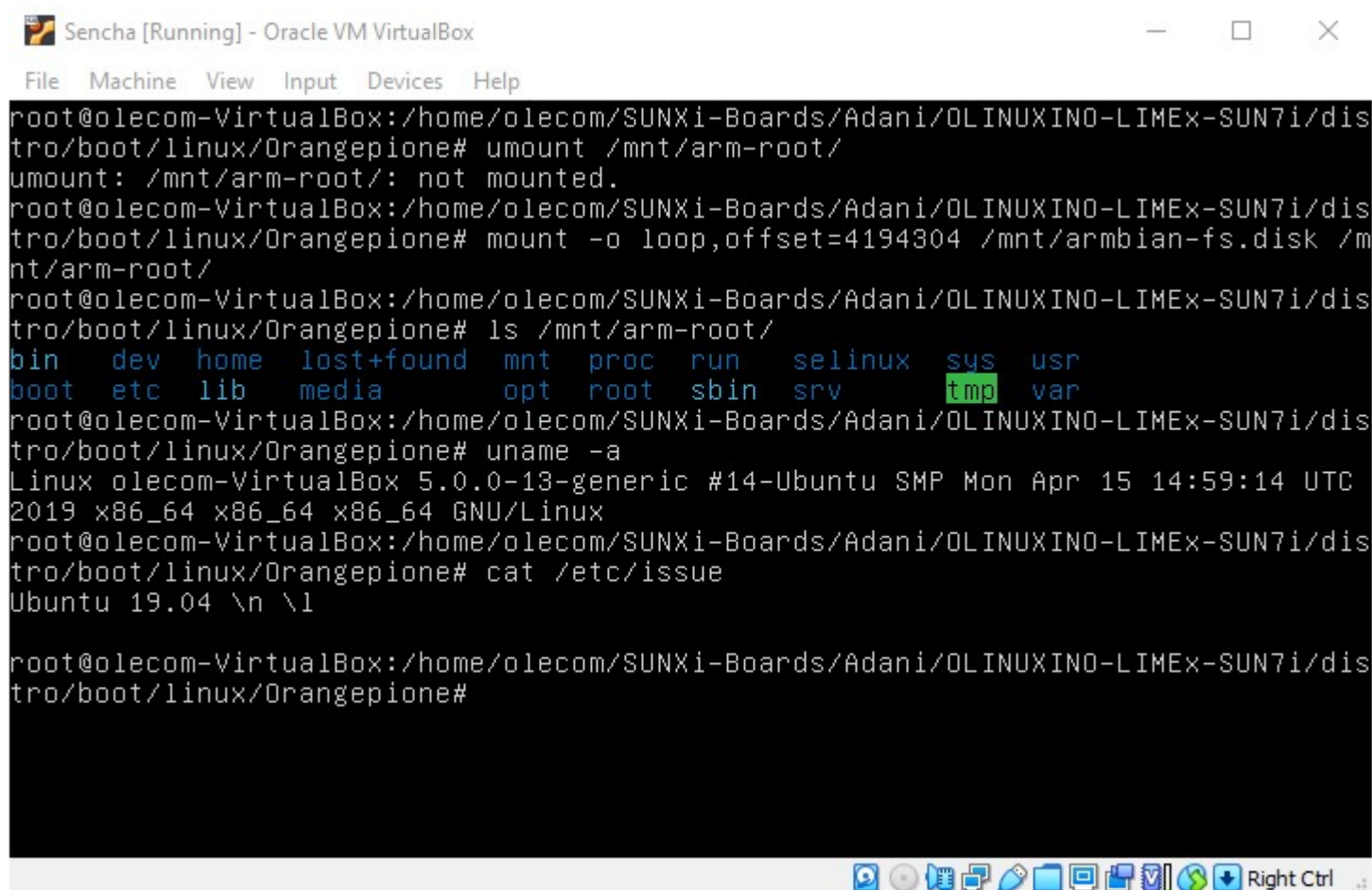


6. qemu

MS Windows 10 + qemu + Armbian for orangepi-pc. Setup description: <https://gist.github.com/wuhanstudio/e9b37b07312a52ceb5973aacf580c453>

Setup is on PC with Linux:

```
xz -d Armbian_22.11.1_Orangepione_jammy_current_5.15.80_minimal.img.xz
fallocate -l 2G /mnt/armbian-fs.disk
dd conv=notrunc of=/mnt/armbian-fs.disk bs=128M if=Armbian_22.11.1_Lime2_bullseye_current_5.15.80.img
mount -o loop,offset=4194304 /mnt/armbian-fs.disk /mnt/arm-root/
# copy kernel, initramfs, dtb files
```



```
Sencha [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@olecom-VirtualBox:/home/olecom/SUNXi-Boards/Adani/OLINUXINO-LIMEEx-SUN7i/distro/boot/linux/Orangepione# umount /mnt/arm-root/
umount: /mnt/arm-root/: not mounted.
root@olecom-VirtualBox:/home/olecom/SUNXi-Boards/Adani/OLINUXINO-LIMEEx-SUN7i/distro/boot/linux/Orangepione# mount -o loop,offset=4194304 /mnt/armbian-fs.disk /mnt/arm-root/
root@olecom-VirtualBox:/home/olecom/SUNXi-Boards/Adani/OLINUXINO-LIMEEx-SUN7i/distro/boot/linux/Orangepione# ls /mnt/arm-root/
bin  dev  home  lost+found  mnt  proc  run  selinux  sys  usr
boot  etc  lib  media  opt  root  sbin  srv  tmp  var
root@olecom-VirtualBox:/home/olecom/SUNXi-Boards/Adani/OLINUXINO-LIMEEx-SUN7i/distro/boot/linux/Orangepione# uname -a
Linux olecom-VirtualBox 5.0.0-13-generic #14-Ubuntu SMP Mon Apr 15 14:59:14 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux
root@olecom-VirtualBox:/home/olecom/SUNXi-Boards/Adani/OLINUXINO-LIMEEx-SUN7i/distro/boot/linux/Orangepione# cat /etc/issue
Ubuntu 19.04 \n \l
root@olecom-VirtualBox:/home/olecom/SUNXi-Boards/Adani/OLINUXINO-LIMEEx-SUN7i/distro/boot/linux/Orangepione#
```

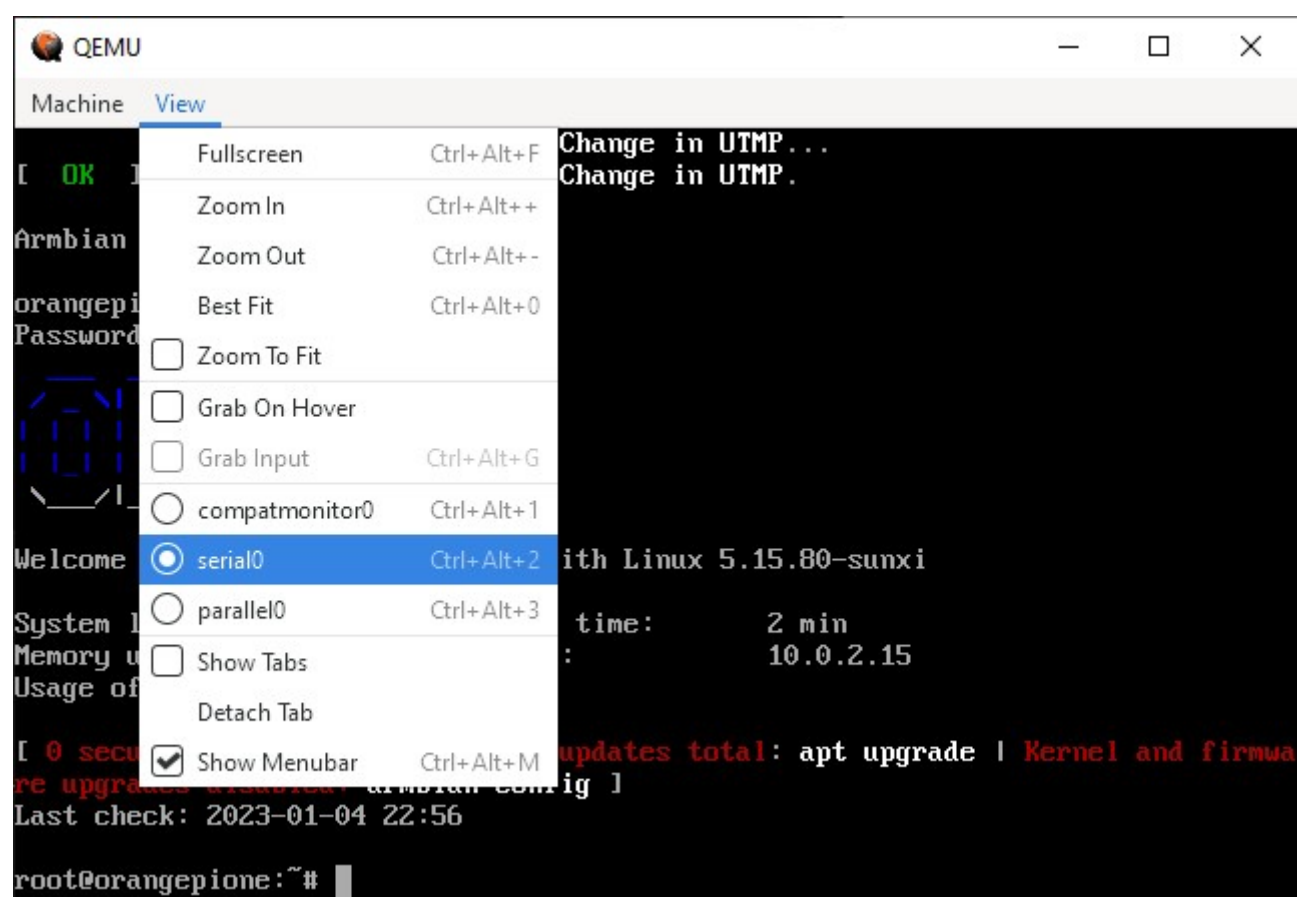
Running on MS Windows (because Linux is too old to install and use **qemu**):

e:_Projects\Adani\OLINUXINO-LIMEEx-SUN7i\distro\boot\linux*.*				
Name	Ext	Size	Date	Attr
[..]		<DIR>	12/26/2022 14:40	----
[Orangepione]		<DIR>	12/26/2022 14:46	----
armbian-fs	disk	2,147,483,648	12/26/2022 15:02	-a--

e:_Projects\Adani\OLINUXINO-LIMEEx-SUN7i\distro\boot\linux\Orangepione*.*				
Name	Ext	Size	Date	Attr
[..]		<DIR>	12/26/2022 14:46	----
initrd.img-5.15	80-sunxi	17,360,672	12/01/2022 00:00	----
sun8i-h3-orangepi-one	dtb	32,065	11/30/2022 14:13	----
ulnitrd-5.15	80-sunxi	17,360,736	12/01/2022 00:00	----
vmlinuz-5.15	80-sunxi	8,393,496	11/30/2022 14:13	----

qemu-system-arm.exe Properties	
General	Compatibility
Security	Details
	Digital Signatures
	Previous Versions
Property	Value
Description	
File description	QEMU machine emulators and tools
Type	Application
File version	7.1.94.0
Product name	QEMU
Product version	7.1.94
Copyright	Copyright various authors. Released un...
Size	15.2 MB
Date modified	12/8/2022 10:34 AM
Language	English (United States)
Legal trademarks	QEMU is a trademark of Fabrice Bellard.

Before that SIGINT handler of current terminal is changed for CTRL+C to CTRL+Q by **stty intr ^Q** thus CTRL+C doesn't kill **qemu** itself. This lets to use CTRL+C as usual process interruption method inside emulation.



Console is OK with input, but **nano** and **mc** are still not working well.